Qatra قطرة

Technical and Strategic Dossier

Blood-donation matching for Algeria

Families ask, associations vouch, donors answer.

Every figure in this document was taken from the repository, from the live database, or from a named public source. Test results are from runs on the date below. Nothing here is aspirational unless it appears in Section 7, which exists to say what is missing.

Yacine Omar Fodhil
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1. Executive summary

Algeria is the best country in Africa at collecting blood and still runs short of it. The Agence Nationale du Sang reports roughly **677 000 bags** from over **800 000 donors**, a rate of **13.98 donors per 1 000 inhabitants** — the highest on the continent for a decade — and describes its own problem not as a shortage of willing people but as *"ce déséquilibre entre l'offre et la demande"*: a timing failure. Only **22% of donors give regularly**, which the agency calls *"relativement faible"*, and its stated request is that citizens **register on regular-donor lists**.

Qatra answers that request. It knows the date every donor becomes eligible again, and it reaches the compatible ones the moment somebody nearby needs their type.

Three things distinguish it from what already exists:

1. **Requests are vouched for.** A verified association in the same wilaya can put its name to a request. The market-leading competitor states in its own store listing that it verifies nothing.
2. **Health data is handled lawfully.** Consent is recorded per purpose with the version of the text shown; a donor's phone number is masked until deliberately opened, and every opening is logged in a record the donor can read and nobody can delete.
3. **The loop closes.** Post, notify, respond, and the family sees who is coming. A registry answers *who could give*. Qatra answers *who is coming, right now*.

2. The problem, with evidence

2.1 Algeria does not have a donor shortage

Figure	Meaning
13.98 per 1 000	Donation rate. First in Africa for around a decade.
677 000 bags	Collected, from 800 000+ donors; up 6.75% since 2023.
22%	**Share of donors who give regularly.** ANS: *"relativement faible"*.
58	Wilayas. Blood is coordinated locally; a donor two provinces away cannot help today.

The ANS names the summer as its pressure point — road-accident season — and has run joint campaigns with the national police and Algérie Télécom to cover it. Its ask of the public is continuity, not volume: *"participer à l'alimentation continue de la banque du sang"*.

2.2 Which means the problem is reach, not recruitment

Seventy-eight per cent of Algerian donors give once and are never effectively called again. Nothing in the existing landscape knows when a given donor became eligible again, or tells them that somebody two kilometres away needs exactly their type this evening. That is the gap this product occupies.

3. Competitive landscape

Install counts and ratings retrieved from Google Play on 24 August 2026. Platform claims verified against the operators' own sites on the same date.

Product	Installs	Rating	Observation
بنك الدم الجزائري (Nabil_Soft)	10K+	4.7 / 130	Market leader. **Contains ads.**
بنك الدم الجزائري (SangDz)	1K+	—	Solo developer based in France.
Sadaka صدقة	1K+	—	—
Nabdz	—	—	Best-designed. 2 requests, 3 donors live.
ONSC / Ministry platforms	—	—	Announced June 2025. No public address.

Approximately **12 000 installs across every blood application in Algeria**, in a country of 45 million with the continent's highest donation rate. The field is not crowded; it is empty.

3.1 The announced national platforms have no front door

The National Observatory of Civil Society launched a donor platform with the Algerian Red Crescent on 14 June 2025; the Ministry of Health announced its own three days later. Fourteen months on:

- `marsad.dz` is the Observatory's institutional site — governance, compliance, tenders. It is actively maintained, with content through April 2026, and carries no donor platform.
- Neither announcement names a URL, an application, or any route by which a citizen reaches the system.
- The Ministry's own site lists five other digital services it does ship — a pharmacy system, a health map, a professional space, a training registry — and no blood platform.

A launch with ministers, a telecom sponsor and no discoverable product is a press event. That may change, and the strategic response is not to compete with it (Section 8).

3.2 Three findings about the market leader

It carries advertising. بنك الدم الجزائري is listed as *Contains ads*. Its developer's other titles are *Easy Hairstyles for Girls*, *PermisDZ*, and several baccalaureate revision apps. Blood donation is the sixth item in a portfolio.

Its data-safety declaration states "No data collected." The application collects, by its own description, a donor's name, wilaya, **blood type** and telephone number. Blood type is health data. That declaration is inaccurate on its face.

It disclaims verification in writing:

> *"This application does not provide any medical services or health advice and does not represent any medical or governmental entity... The information provided is provided by the users."*

The fraud problem is therefore not an unrecognised gap in the market. It is a term of service.

3.3 And the most serious competitor demonstrates the same emptiness

Nabdz is the best-built of them — 58 wilayas, urgency tiers, a pledge mechanic comparable to Qatra's response counter. Its live public grid currently shows two requests and three donors, one of the requests titled **"Test Patient"**.

4. Regulatory position

This section states the reasoning behind decisions already implemented in the codebase. It is engineering rationale, not legal advice, and the open question in Section 4.4 should be settled with counsel or with the ANS before any public launch.

4.1 Health data — Loi 18-07 and Loi 25-11

A patient's name attached to a blood type, and a donor's telephone number attached to theirs, are health data. Qatra treats them accordingly:

- **Consent is recorded per purpose, with the version of the text shown.** Two purposes exist — `health_data` and `contact_sharing` — each stamped with the wording the user actually read (`health-data-v1`, `contact-sharing-v1`). Consent given against superseded wording stops granting access until the user agrees again.
- **Consent records have no delete policy.** They are the evidence that processing was lawful, and evidence a controller can erase is not evidence.
- **Donor numbers are masked until deliberately opened.** Search returns `05 •• •• •• 56`. A single function returns the whole number, and it writes a row while doing so.
- **Every reveal is logged, and the donor can read the log.** The record names who opened the number, for which donor, on behalf of which association, and when. It cannot be updated or deleted by anyone.
- **Data-subject rights have a route in the product** — export, correction, deletion — rather than an email address on a policy page.

4.2 Who may vouch — Loi 12-06 on associations

Verification publishes an attestation under an association's name. An association is a legal person acting through its statutory representatives, not through whichever member happens to be holding a phone; if a vouched request proves fraudulent, the association carries it. Verification is therefore restricted to association *administrators* in the request's own wilaya, enforced in the database rather than the interface.

Volunteers keep everything else — the request list, donor search, the work of mobilising people. They simply cannot sign the association's name.

"Administrator" is a proxy for "entitled to bind the association". A real committee's mandate structure may be richer, and blood-request vetting is not itself a regulated medical act. This mapping is marked in the migration and should be confirmed with the ANS.

4.3 No payment, ever

Blood donation in Algeria is voluntary and unpaid. Paying donors — in cash, vouchers, credit, priority, or anything exchangeable for them — compromises the safety of the supply, because a donor with a financial motive has a reason to conceal a disqualifying history.

This is a documented invariant at the top of the compliance module and a code-review gate, not a preference. Non-transferable recognition (streaks, certificates) is permitted precisely because it cannot be exchanged.

4.4 Open question: data residency

The production database is hosted in AWS `us-east-1`. Algerian patients' names, blood types and telephone numbers therefore rest on servers outside Algeria. Four files in the repository carry a `TODO(compliance)` marker recording this.

This is a decision to be taken deliberately, and the options are known: a European region shortens the answer, and domestic hosting removes the question entirely. It is a launch gate, not a demonstration gate, but it is the question a well-informed panel will ask.

5. How the loop closes

1. **A family posts a request.** Three steps: who needs blood, where, how urgent. The hospital is free text — a family knows their hospital by name, not by an identifier in a directory.
2. **Phone verification interrupts *after* the form, not before.** A plea written at 3am is captured while the person still has the words; the draft is held and posted the moment the number is confirmed. It is never asked for twice.
3. **Compatible donors nearby are notified.** Same wilaya, compatible blood type, eligible now — not inside the 90-day cooldown — and never the person who posted it.
4. **A committee may vouch for it.** The console groups requests by what to do next: needs review, open a long time, already vouched.
5. **A donor responds, and the family sees it.** The count is public so twenty people do not arrive for two units; the identities are not.
6. **A donor who cannot go says so.** Withdrawal is a state change the family sees, not an erasure.

6. Technical architecture

6.1 Stack

Web application	React 18, Vite 6, TypeScript, Tailwind for layout
Mobile shells	Capacitor (Android, iOS) over the same source
Backend	Supabase: PostgreSQL, PostgREST, GoTrue, Edge Functions (Deno)
Notifications	Web Push (VAPID), queued in the database, sent by a worker
Maps	Leaflet over OpenStreetMap
Languages	English, French, Arabic, with full right-to-left support

The core package is framework-agnostic by design: configuration is injected rather than read from the bundler's environment, so a non-Vite host can consume the same logic.

6.2 Security model

Authorisation lives in the database, not the interface. **54 row-level security policies** and **23 `SECURITY DEFINER` functions** decide who may read and write what; the client can only ask.

Three principles are worth stating because they were arrived at by correcting mistakes:

- **One right per predicate.** A single function once gated vouching, patient-record access *and* donor search. Narrowing it for one purpose silently removed a right unrelated to the decision. They are now separate.
- **Two layers, not one.** Revoking a function from `public` does not remove Supabase's separate grants to the `anon` and `authenticated` roles. Several functions were therefore reachable by anonymous callers and stopped only by their own internal guard. The test suite now asserts the grants themselves.
- **Logs that cannot be rewritten.** Consent records and contact reveals have no update or delete policy at all.

6.3 Notifications

Database triggers write to an outbox; an edge function drains it. An HTTP call inside the transaction would make *posting a plea for blood* fail whenever a push service was slow, and a failed send would leave nothing to retry.

Each claim takes a five-minute lease. `SKIP LOCKED` alone separates only simultaneous workers — a second worker a moment later would send the same notification again, and the test suite caught exactly that.

The payload carries blood type and wilaya and nothing else. A push lands on a lock screen that anyone nearby can read, so it says only what the public request list already shows.

6.4 Blood compatibility

The full eight-by-eight red-cell table is written out rather than derived, so a reader can check it against a transfusion chart line by line. Unknown or missing types answer *false*, never *true*: the safe answer to "do we know this is safe?" is no.

The unit tests check the table from both directions against an independently written chart, verify that rhesus is not symmetric, and confirm that a donor whose type is unrecorded is asked rather than refused.

6.5 Verification and testing

Suite	Count	What it covers
`verify:db`	188	RLS against real PostgreSQL, as several users
`test:unit`	14	Blood compatibility both directions; pledge arithmetic
Playwright	32	End-to-end journeys, three languages, two viewports
Migrations	26	Applied in filename order
Screens	24	
Components	41	
Translated strings	387	Each in all three languages, enforced by the type system

A missing translation is a compilation error, not a visible gap: the string table is a TypeScript interface, so a key added in one language and forgotten in another fails the build.

7. What is built, and what is not

State	Capability
Built	Patient/family posts a request; association verifies it
Built	Donor discovery, map, urgency and recency ordering
Built	Donor responds; family sees the count; withdrawal supported
Built	Web push: compatible donors in-wilaya, and the family on a response
Built	Consent, contact-reveal logging, data-subject requests
Built	Blood compatibility with tests
Built	Trilingual EN / FR / AR with right-to-left
Built	Committee invite links: a committee brings the donors it already has
Built	Public demonstration deployment
Open	Demonstration OTP is enabled: any number can be claimed
Open	Data residency (Section 4.4)
Open	"Administrator" as proxy for authority to bind an association
Not built	Native push for the Capacitor Android build (needs FCM)
Not built	Android build refreshed against the current application
Not built	Pagination beyond a 200-request cap per wilaya
Not built	Any integration with a national registry or hospital system

8. Positioning

8.1 Not "first"

There is a national registry announced and at least three applications live. Claiming to be first is both false and checkable in thirty seconds, and volunteering the competition first reads as command where being caught by it reads as naivety.

8.2 The line

> *Algeria has a national donor registry. It answers* who could give. *Qatra answers* who is coming, right now, and can this request be trusted — *and it does so to a standard a committee can put its name on.*

8.3 Why the Red Crescent partnership strengthens

The Algerian Red Crescent is already the state's chosen partner in this space, having co-launched the June 2025 platform. A CRA wilaya committee vouching through Qatra is therefore not a rival to the national effort but the local layer it does not have. The compliance architecture is what makes that partnership possible: an institution can only lend its name to a system that handles health data properly.

9. Sources

- Agence Nationale du Sang, on regular-donor lists and the summer supply–demand imbalance: `https://news.radioalgerie.dz/fr/node/47717`
- ANS collection figures and donation rate: `https://www.algerie360.com/don-de-sang-lagence-nationale-donne-`
- ONSC platform launch: `https://www.dzair-tube.dz/en/algeria-launches-first-digital-blood-donor-platfo`
- Ministry of Health platform: `https://meatechwatch.com/2025/06/18/algeria-launches-digital-platform-to-e`
- National Observatory of Civil Society: `https://marsad.dz`
- Ministry of Health: `https://www.sante.gov.dz`
- بنك الدم الجزائري (Nabil_Soft) on Google Play: `https://play.google.com/store/apps/details?id=com.dz.bank.blood.nabilsoft.bankblooddz`
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